

MASTER INTAKE — 12-SECTION PROJECT AUDIT

Project: Quick Analysis

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Model: STRUCTMIND CORE · PRO

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1. FILE INVENTORY & DRAWING STATUS

Drawing Register

#	File / Sheet Name	Discipline	Rev	Status	Notes
1	S0.01	Structural	—	■ Readable	General Notes
2	S0.02	Structural	—	■ Readable	Typical Schedules & Details
3	S0.03	Structural	—	■ Readable	Special Inspections
4	S1.01	Structural	—	■ Readable	Foundation Plan
5	S1.02	Structural	—	■ Readable	Second Floor Framing Plan
6	S1.03	Structural	—	■ Readable	Typical Level Framing Plan (Reference)
7	S1.04	Structural	—	■ Readable	Roof Framing Plan

#	File / Sheet Name	Discipline	Rev	Status	Notes
8	S2.01	Structural	—	■ Readable	Structural Wall Designation Plan
9	S2.02	Structural	—	■ Readable	CMU and Wood Stud Framing Schedules
10	S2.03	Structural	—	■ Readable	CMU Reinforcement Elevations - Stair #1/Elevator
11	S2.04	Structural	—	■ Readable	CMU Reinforcement Elevations - Stair #2
12	S3.01	Structural	—	■ Readable	Foundation Sections
13	S3.02	Structural	—	■ Readable	Floor Framing Sections
14	S3.03	Structural	—	■ Readable	Roof Framing Sections
15	S3.04	Structural	—	■ Readable	Roof Framing Sections
16	S3.05	Structural	—	■ Readable	Canopy Framing Sections
17	S3.06	Structural	—	■ Readable	Trellis and Site Wall Details

- Status: ■ Readable · ■■ Partial · ■ Unreadable · ■ Referenced but Missing
- List every uploaded file — never group

Package Statistics

- Total files uploaded: 17
- Structural sheets: 17 | Architectural: 0 | Civil: 0 | Other: 0
- Unreadable / scanned: None
- Sheets referenced but not uploaded: Architectural drawings (for opening widths, canopy details, trellis layout)
- Recommended action before detailing:
 - ■ Obtain Architectural drawings for cross-referencing steel lintel lengths and canopy/trellis geometry.
 - ■ Issue RFIs for all missing critical dimensions (e.g., column lengths).

2. PROJECT IDENTITY & SYSTEM SUMMARY

Project Identity Table

Field	Extracted Value	Source Sheet	Status
Project Name	HOME 2 - CAROWINDS	S0.01	■ Found
Project Number	2024.023	S0.01	■ Found
Location / Address	Cabelas Dr, Fort Mill, South Carolina 29708	S0.01	■ Found
Owner	Samarth Patel / Neelkanth Hospitality	S0.01	■ Found
General Contractor	NOT FOUND IN PROVIDED FILES	—	■ Not applicable / Not found
EOR (Structural Engineer)	Taylor & Viola Structural Engineers	S0.01	■ Found
Architect	rba.	S0.01	■ Found

Field	Extracted Value	Source Sheet	Status
Fabricator (if noted)	NOT FOUND IN PROVIDED FILES	—	■ Not applicable / Not found
Approval Stage	CONSTRUCTION DOCUMENTS	S0.01	■ Found
Code of Record	IBC 2018 / AISC 360-16	S0.01	■ Found
Seismic Design Category	SDC C	S0.01	■ Found
Wind Exposure Category	C	S0.01	■ Found

Structural System Summary

- Primary system: Wood-framed structure with CMU shear walls and stair/elevator cores.
- Lateral system: CMU shear walls and wood shear walls.
- Floor system: Engineered wood I-joists with 3/4" T&G; sheathing.
- Roof system: Pre-engineered wood trusses with 3/4" T&G; sheathing.
- Foundation interface: Concrete shallow foundations (spread footings and wall footings).
- Special conditions: Very limited structural steel scope consisting of columns, beams, and lintels supporting wood framing. Outdoor trellis structure.

Approximate Tonnage Estimate

- Primary structural steel: ~4.5 t
- Secondary / misc steel: ~0.8 t
- Combined project total: ~5.3 t
- Basis: "Counted from framing plan S1.02 and schedules S0.02. Tonnage is highly preliminary as all column and lintel lengths are missing and must be assumed or scaled."

3. GRID & GEOMETRY AUDIT

Grid Line Inventory

Grid	Direction	Spacing	Sheet	Consistent Across Sheets?	Issues
1 to 18	E-W	12'-10" typical	S1.01, S1.02	■ Yes	None
A to G	N-S	Varies (e.g. 21'-4", 35'-10")	S1.01, S1.02	■ Yes	None

- Grid origin confirmed: No
- Skew or angle grids: No
- Sloped / cambered members: No
- Curved geometry: No
- Grid conflicts between sheets: None detected

4. MATERIAL GRADE NORMALIZATION

All Materials Found

Member Category	Raw Callout	Normalized ASTM Grade	Source Sheet	Conflict?	Notes
W-Shapes	FY=50 KSI	A992 Gr.50	S0.01	■ No	Standard for W-shapes.
HSS Shapes	ASTM A500 GRADE C	A500 Gr.C	S0.01	■ No	Standard for HSS.
Angles	FY=36 KSI	A36	S0.01	■ No	L-shapes for lintels.
Plates	FY=36 KSI	A36	S0.01	■ No	Baseplates, cap plates, lintel plates.
Anchor Rods	ASTM F1554, GR. 36	F1554 Gr.36	S0.01	■ No	—

Member Category	Raw Callout	Normalized ASTM Grade	Source Sheet	Conflict?	Notes
High Strength Bolts	ASTM A325N	A325-N	S0.01	■ No	N-type (threads in shear plane).
Weld Filler Metal	70 KSI MIN TENSILE	E70XX	S0.01	■ No	Standard.
Galvanizing	GALVANIZED	ASTM A123 / A153	S0.01	■ No	Callout for bearing plates on S0.02.

Normalized Grade Summary

Category	Grade	Source	Status
W-shapes	A992 Gr.50	S0.01	■ OK
HSS / Tube	A500 Gr.C	S0.01	■ OK
Plates	A36	S0.01	■ OK
Anchor bolts	F1554 Gr.36	S0.01	■ OK
Structural bolts	A325-N	S0.01	■ OK
Weld filler metal	E70XX	S0.01	■ OK

5. SCOPE DETECTION & CLASSIFICATION

Member Scope Register

#	Member Type	Detected?	Qty (approx)	In Fab Scope	Source Sheet	Notes
1	Columns(W)	■ Not Found	0	■ NO	—	—

#	Member Type	Detected?	Qty (approx)	In Fab Scope	Source Sheet	Notes
2	Columns(HSS)	■ Yes	10	■ YES	S0.02, S1.01, S3.06	Main bldg columns and trellis columns.
3	Primary Beams	■ Yes	8	■ YES	S1.02, S3.06	W16, HSS18x6, HSS8x6 members.
4	Secondary Beams	■ Not Found	0	■ NO	—	—
5	Transfer Beams	■ Not Found	0	■ NO	—	—
6	Purlins	■ Not Found	0	■ NO	—	Wood scope.
7	Girts	■ Not Found	0	■ NO	—	Wood scope.
8	Roof Joists	■ Not Found	0	■ NO	—	Wood trusses.
9	Floor Joists	■ Not Found	0	■ NO	—	Wood I-joists.
10	Vertical Bracing(HSS)	■ Not Found	0	■ NO	—	—
11	Horiz. Bracing	■ Not Found	0	■ NO	—	—
12	Moment Frames	■ Not Found	0	■ NO	—	—
13	Shear Plates	■ Not Found	0	■ NO	—	—

#	Member Type	Detected?	Qty (approx)	In Fab Scope	Source Sheet	Notes
14	Trusses	■ Not Found	0	■ NO	—	Wood scope.
15	Space Frames	■ Not Found	0	■ NO	—	—
16	Stairs	■ Uncertain	0	■ UNCLEAR	—	Stairs shown but construction not defined.
17	Landings	■ Uncertain	0	■ UNCLEAR	—	Assumed concrete or wood.
18	Handrails/Guardrails	■ Uncertain	0	■ UNCLEAR	—	Assumed Arch. scope.
19	Ladders	■ Not Found	0	■ NO	—	—
20	Mezzanines	■ Not Found	0	■ NO	—	—
21	Platforms/Walkways	■ Not Found	0	■ NO	—	—
22	Canopies	■ Yes	1	■ YES	S1.02	Pre-fabricated by others. Our scope TBD.
23	Shade Structures	■ Yes	1	■ YES	S3.06	Trellis structure.
24	Bollards	■ Not Found	0	■ NO	—	—
25	Gates	■ Not Found	0	■ NO	—	—

#	Member Type	Detected?	Qty (approx)	In Fab Scope	Source Sheet	Notes
26	Fences	■ Not Found	0	■ NO	—	—
27	Site Rails	■ Not Found	0	■ NO	—	—
28	Embeds	■ Not Found	0	■ NO	—	—
29	Crane Beams	■ Not Found	0	■ NO	—	—
30	Crane Rails	■ Not Found	0	■ NO	—	—
31	Base Plates (Loose)	■ Yes	10	■ YES	S0.02, S3.06	For all HSS columns.
32	Base Plates (Assembly)	■ Yes	10	■ YES	S0.02, S3.06	—
33	Anchor Bolt Plans	■ Yes	1	■ YES	S1.01	AB locations shown, details are incomplete.
34	Precast Interface Steel	■ Not Found	0	■ NO	—	—
35	Galvanizing (noted)	■ Yes	—	■ YES	S0.02	Bearing plates noted as galvanized.
36	Misc Plates/Clips/Angles	■ Yes	—	■ YES	S0.02	Lintel angles, lintel plates, stiffeners.

#	Member Type	Detected?	Qty (approx)	In Fab Scope	Source Sheet	Notes
37	Delegated Connection Design	■ Yes	—	■ YES	S0.01	General Note 12 explicitly delegates.
38	CNC / NC Files Required	■ Yes	—	■ YES	—	Standard deliverable.

- Items confirmed IN fabrication scope: HSS Columns, W-shape and HSS beams, steel lintels (W, L), base plates, miscellaneous plates, trellis structure steel.
- Items confirmed OUT of scope: Wood framing (joists, trusses, studs), CMU walls.
- Items requiring scope clarification RFI:
- "Pre-fabricated Canopy by Others" (S1.02): Clarify if any structural steel support framing is in our scope.
- Stairs: Confirm material and if structural steel stringers/treads are required.

6. ANCHOR BOLT & BASEPLATE INTAKE

Anchor Bolt & Baseplate Schedule

Col Mark	Pattern	Bolt Dia	Spec/Grade	Embed	Projection	BP Size	BP Thk	Grout Thk	Hole Type	Sheet	Status
C5	4-Bolt	3/4"	F1554 Gr.36	■ MIS SING	■ MIS SING	12"x 12"	3/4"	■ MIS SING	STD	S0.02	■ Fail
C6	4-Bolt	3/4"	F1554 Gr.36	■ MIS SING	■ MIS SING	12"x 12"	1"	■ MIS SING	STD	S0.02	■ Fail
C8B	4-Bolt	1"	F1554 Gr.36	■ MIS SING	■ MIS SING	16"x 16"	1"	■ MIS SING	STD	S0.02	■ Fail
Trellis Col	4-Bolt	1"	F1554 Gr.36	12"	■ MIS SING	16"x 16"	1"	■ MIS SING	STD	S3.06	■■ Warning

7. CONNECTION INTELLIGENCE

Connection Assumption Register

Joint Location	Members Connected	Connection Type	Bolt Size/Grade	Weld Size/Type	Plate Thickness	Slip-Critical?	Deferred?	RFI?
Beam-to-CMU	W16x36 to CMU Wall	Bearing Plate	N/A	Field	3/8"	■ No	■ Yes	■ Minor
Beam-to-Column	HSS18x6 to HSS Column	HSS Shear Tab	3/4" A325-N	5/16" Fillet	3/8"	■ No	■ Yes	■ Major
Beam-to-Beam	W14x22 to HS S18x6	Simple Shear Tab	3/4" A325-N	5/16" Fillet	3/8"	■ No	■ Yes	■ Major
Column Base	HSS Column to FDN	Base Plate	See Sec. 6	5/16" Fillet	See Sec. 6	■ No	■ Yes	■ Critical
Wind Post Top	Steel Beam to HSS Post	Thru-Bolted	(2) 1/2"	None	1/2"	■ No	■ No	■ No

> ■■ Slip-Critical Alert (complete if SC connections found): > | Check | Status | > | :--- | :---: | > | SSPC surface prep spec stated? | ■ Not applicable / Not found | > | Faying surface masking noted? | ■ Not applicable / Not found |

| Bolt pre-tension method stated? | ■ Not applicable / Not found | > | Surface Class (A/B) confirmed? | ■ Not applicable / Not found |

8. SPECIFICATION CONFLICT MATRIX

Conflicts Detected

ID	Item	Structural Callout	Arch/Other Callout	Conflict Type	Impact	Resolution
—	—	—	—	—	—	No conflicts detected within the provided structural drawing set.

9. PRELIMINARY MTO (MATERIAL TAKE-OFF)

Complete MTO Register

#	Type	Mark	Profile	Qty	Length (Imperial)	Length (mm)	Unit Wt (kg/m)	Wt (kg)	Wt (lb)	Grade	Sheet	Confidence
1	Beam	—	HSS18X6X3/8	6	21'-4"	6,502	NF-WT	0	0	A500-C	S1.02	LOW
2	Beam	—	W16X36	1	21'-4"	6,502	53.6	349	769	A992	S1.02	MEDIUM
3	Beam	—	W14X22	8	12'-10"	3,912	32.7	1,023	2,256	A992	S1.02	MEDIUM
4	Header	—	HSS9X5X3/8	1	12'-10"	3,912	NF-WT	0	0	A500-C	S1.02	LOW

#	Type	Mark	Profile	Qty	Length (Imperial)	Length (mm)	Unit Wt (kg/m)	Wt (kg)	Wt (lb)	Grade	Sheet	Confidence
5	Lintel	L1	W8X24	—	NF-L EN	0	35.7	0	0	A992	S0.02	LOW
6	Lintel	L2	W16X50	—	NF-L EN	0	74.4	0	0	A992	S0.02	LOW
7	Column	C5	HSS6X6 X5/16	1	NF-L EN	0	NF-WT	0	0	A500-C	S0.02	LOW
8	Column	C6	HSS6X6 X3/8	2	NF-L EN	0	33.1	0	0	A500-C	S0.02	LOW
9	Column	C8B	HSS8X6 X3/8	1	NF-L EN	0	NF-WT	0	0	A500-C	S0.02	LOW
10	Trellis Bm	—	HSS8X6 X3/8	4	Est. 13'-0"	3,962	NF-WT	0	0	A500-C	S3.06	LOW
11	Trellis Col	—	HSS8X6 X3/8	6	Est. 8'-0"	2,438	NF-WT	0	0	A500-C	S3.06	LOW
12	Plate	BP-C5	PL 3/4"	1	1'-0" x 1'-0"	305x305	—	14	31	A36	S0.02	HIGH
13	Plate	BP-C6	PL 1"	2	1'-0" x 1'-0"	305x305	—	37	82	A36	S0.02	HIGH

#	Type	Mark	Profile	Qty	Length (Imperial)	Length (mm)	Unit Wt (kg/m)	Wt (kg)	Wt (lb)	Grade	Sheet	Confidence
14	Plate	BP-C8B	PL 1"	1	1'-4" x 1'-4"	406x406	—	33	72	A36	S0.02	HIGH
15	Plate	BP-Trellis	PL 1"	6	1'-4" x 1'-4"	406x406	—	196	433	A36	S3.06	HIGH

MTO Summary by Category

Category	Count	Total Wt (lb)	Total Wt (kg)	Total (tons)
W-Shapes	10	3,025	1,372	1.51
HSS / Tube	20	0	0	0.00
Pipe	0	0	0	0.00
Angles / Channels	0	0	0	0.00
Plates	10	618	280	0.31
Misc / Anchors	0	0	0	0.00
PROJECT TOTAL	40	3,643	1,652	1.82

Note: Total tonnage is artificially low due to missing HSS unit weights and critical missing lengths for all columns and lintels. Actual tonnage is likely 3-4x higher.

10. DRAWING QUALITY SCORE

Quality Assessment

Indicator	Score (1–5)	Finding	Blocking?
Revision / Approval Stage	4	Issued as "Construction Documents". Clear but not yet "For Fabrication".	No
Connection Design Completeness	1	Fully delegated. No prescriptive details provided for key connections.	Yes
Dimensional Clarity	2	Grid is clear, but all column lengths and many beam/lintel lengths are missing.	Yes
Scope Definition	3	Steel scope is identifiable but requires RFI for clarification on canopy/stairs.	No
Specification Availability	5	General notes are comprehensive and clear for materials and codes.	No
Cross-Sheet Consistency	4	No conflicts found between plan, schedule, and notes.	No
AISC / AWS Compliance Indicators	3	Standard codes are referenced, but detailing is absent.	Yes
OVERALL SCORE	22/35	Grade: B — Significant missing information prevents fabrication.	—

Fabrication Start Recommendation: ■ HOLD Reason: Critical dimensions (all column lengths, lintel lengths) and anchor bolt details (projection, embedment) are missing, which blocks material procurement and model creation.

11. ISSUE REGISTER

All Blocking + Non-Blocking Issues

ID	Priority	Type	Description	Sheet	Blocks Fab?	Suggested RFI
1	■ Critical	MISSING-DIM	All column lengths are missing from the Steel Column Schedule.	S0.02	■ Yes	RFI-001
2	■ Critical	MISSING-DIM	Anchor bolt embedment and projection dimensions are missing for all main building columns (C5, C6, C8B).	S0.02	■ Yes	RFI-002
3	■ Major	MISSING-DIM	Lengths for steel lintels L1, L2, and loose angle lintels are not provided. Schedule refers to Arch. drawings.	S0.02	■ Yes	RFI-003

ID	Priority	Type	Description	Sheet	Block's Fab?	Suggested RFI
4	■ Major	CONNECTION-INCOMPLETE	Connection design is delegated. Please confirm required submittals (calculations, erection drawings, etc.) and any load information or special considerations for the delegated engineer.	S0.01	■ Yes	RFI-004
5	■ Minor	SCOPE-GAP	Plan S1.02 notes "PRE-FABRICATED CANOPY BY OTHERS". Please clarify if any structural steel supports, posts, or connections to the main structure are within the steel fabricator's scope.	S1.02	■ No	RFI-005
6	■ Minor	MISSING-DETAIL	Grout thickness under base plates is not specified. Please provide required thickness.	S0.02	■ No	RFI-006

ID	Priority	Type	Description	Sheet	Block s Fab?	Suggested RFI
7	■ Observation only	NF-WT	Unit weight for HSS18x6x3/8, HSS9x5x3/8, HSS6x6x5/16, HSS8x6x3/8 not found in standard AISC tables. Please confirm availability or provide substitution.	S1.02, S0.02	■ No	RFI-007

Summary: 2 Critical · 2 Major · 2 Minor · 1 Observation only · Total: 7 issues

12. READY-TO-SEND RFI PACKAGE

■ Critical RFIs (must answer before modeling): RFI-001 through RFI-002 --- RFI-001 To: Taylor & Viola Structural Engineers Re: S0.02 — Steel Column Schedule Priority: Critical Blocking: Yes

Question: The Steel Column Schedule on sheet S0.02 does not list the lengths for column marks C5, C6, and C8B. Please provide a complete schedule with all required column lengths (or top/bottom of steel elevations) to allow for material procurement and modeling.

Expected Response Format: A revised Steel Column Schedule on sheet S0.02 or a supplementary schedule providing all column lengths.

RFI-002 To: Taylor & Viola Structural Engineers Re: S0.02, S1.01 — Anchor Bolt Details for Main Columns Priority: Critical Blocking: Yes

Question: The Steel Column Schedule on S0.02 and foundation details do not specify the required embedment depth or above-concrete projection for the anchor bolts on columns C5, C6, and C8B. Please provide these critical dimensions.

Expected Response Format: Written confirmation of dimensions or a detail showing required embedment and projection for each column type.

--- ■ Urgent RFIs (answer within first week): RFI-003 through RFI-004 --- RFI-003 To: Taylor & Viola Structural Engineers Re: S0.02 — Lintel Schedules Priority: Urgent Blocking: Yes

Question: The Lintel Schedule and Brick Veneer Loose Lintel Schedule on sheet S0.02 do not provide lengths for any of the steel lintels (e.g., L1, L2, L7x4x3/8). The schedules refer to opening sizes on

Architectural drawings which were not included in the bid package. Please provide all steel lintel lengths or the relevant architectural drawings.

Expected Response Format: A complete lintel schedule with lengths, or issuance of the corresponding architectural elevation/plan sheets.

RFI-004 To: Taylor & Viola Structural Engineers Re: S0.01 — General Structural Note #12 Priority: Urgent Blocking: Yes

Question: General Structural Note #12 delegates the design of steel connections to the fabricator. Please confirm the required submittal package for this delegated design (e.g., sealed calculations, erection drawings with connection details). Additionally, please provide any specific connection load tables, criteria, or special considerations that must be used for the design.

Expected Response Format: Written confirmation of delegated design submittal requirements and provision of any necessary design criteria or load tables.

--- ■ Standard RFIs (answer before drawing release): RFI-005 through RFI-007 --- RFI-005 To: Taylor & Viola Structural Engineers; rba. Re: S1.02 — Canopy Scope Priority: Standard Blocking: No

Question: The Second Floor Framing Plan (S1.02) indicates a "PRE-FABRICATED CANOPY BY OTHERS". Please clarify if the structural steel fabricator's scope includes any support beams, columns, embeds, or connection hardware associated with this canopy.

Expected Response Format: Written clarification of the steel fabrication scope related to the canopy.

RFI-006 To: Taylor & Viola Structural Engineers Re: S0.02 — Base Plate Grout Thickness Priority: Standard Blocking: No

Question: The base plate details shown in the Steel Column Schedule (S0.02) do not specify a grout thickness. Please provide the required grout thickness to be used under all steel column base plates.

Expected Response Format: Written confirmation of the required grout thickness.

RFI-007 To: Taylor & Viola Structural Engineers Re: S0.02, S1.02 — Non-Standard HSS Sizes Priority: Standard Blocking: No

Question: The drawings call for several HSS sizes (HSS18x6x3/8, HSS9x5x3/8, HSS6x6x5/16, HSS8x6x3/8) that are not common production sizes per AISC. Please confirm these sections are available or provide preferred substitute sections that are readily available.

Expected Response Format: Confirmation of availability or a list of acceptable substitute sections.

ACTION SUMMARY

Priority Actions for This Analysis

#	Priority	Action Required	Owner	Deadline	Blocking?
1	■ Critical	Issue RFI-001 for missing column lengths.	Estimator/PM	ASAP	■ Yes
2	■ Critical	Issue RFI-002 for missing anchor bolt embed/projection.	Estimator/PM	ASAP	■ Yes
3	■ Major	Issue RFI-003 for missing lintel lengths.	Estimator/PM	ASAP	■ Yes
4	■ Major	Issue RFI-004 to clarify delegated design submittal requirements.	Estimator/PM	ASAP	■ Yes
5	■ Minor	Request architectural drawings for context and verification.	Estimator/PM	ASAP	■ No